

Leveraging Large Language Models for Personalized Parkinson's Disease Treatment

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Abstract—The treatment of Parkinson's Disease (PD) remains a major clinical challenge in modern medicine due to the lack of a definitive cure and the complexity of managing heterogeneous symptoms. Patients often require lifelong medication, necessitating physicians create individualized treatment plans to address diverse symptom profiles. However, traditional approaches to medication management are often limited by factors such as high costs and the limited number of patients a doctor can attend to. Although deep learning techniques are applied in PD treatment, they typically lack interpretability and are confined to numerical data inputs. In this study, we propose a novel framework that leverages large language models (LLMs) to design personalized treatment strategies for PD. Our approach integrates patient information in natural language form and external textual knowledge, such as medical guidelines, to generate personalized prescriptions. To enhance effectiveness, we use Monte Carlo Tree Search (MCTS) to refine strategies and establish a robust medication recommendation dataset. To enhance reliability and interpretability, we incorporate Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) reasoning within the LLM system, ensuring that each proposed strategy is accompanied by step-by-step explanations and references to similar historical cases. We validate our system using the Parkinson's Progression Marking Initiative (PPMI) dataset, featuring longitudinal patient records that ensure a statistically robust evaluation of our approach. Experimental results show that, compared to DL models, our method surpasses human-physician performance, reducing Unified Parkinson's Disease Rating Scale (UPDRS) scores by over 1.4 points on average. Furthermore, over 43% of patients achieve clinically significant improvements. A detailed case study underscores the flexibility of LLM in dynamically adjusting medication plans for patients at different disease stages, highlighting its potential to advance personalized PD management in real-world settings.

Index Terms—Parkinson's Disease, LLM, treatment strategies, personalized prescription

I. INTRODUCTION

Parkinson's Disease (PD) poses a major challenge that exacerbates public health pressures. According to the World Health Organization (WHO) [1], approximately 7 to 10 million individuals are currently living with Parkinson's Disease

globally, the majority of whom are elderly. As the global aging population continues to rise, the number of Parkinson's Disease patients will increase to between 12 and 17 million by 2040 [2].

Despite significant advances in medical science, PD remains an incurable neurodegenerative disorder. Traditional treatment strategies primarily aim to alleviate symptoms rather than halt or reverse the progression of the disease. Levodopa, the most commonly prescribed medication, can temporarily improve motor symptoms by replenishing dopamine levels in the brain. However, due to the complex drug-release mechanism of Levodopa, physicians often need to adjust dosages through trial and error, a process that is not only time-consuming but also prone to imprecision. Traditional medication regimens are heavily reliant on the expertise of neurologists, who are tasked with the continuous monitoring and individualized adjustment of drug therapies based on the patient's symptoms. This results in high treatment costs and frequent patient visits, further placing a strain on healthcare systems.

Moreover, it is reported that the number of neurologists is insufficient to meet the growing demand for PD care [3], meaning that many patients are unable to receive timely and effective treatment. Neurodegenerative diseases, including Parkinson's Disease, are anticipated to present significant challenges to public health in the coming decades, necessitating the development of more intelligent and effective management strategies to mitigate the growing burden on healthcare systems.

Recent advancements in deep learning have enabled the exploration of AI to assist in healthcare and the treatment of Parkinson's Disease. While AI offers a promising approach to enhancing treatment strategies, three critical challenges remain: **effectiveness, reliability, and interpretability**. First, deep learning systems that rely solely on numerical data while neglecting textual information are inherently constrained in their ability to fully process and comprehend the disease. Second, the reliability of AI systems directly impacts both patient safety and therapeutic efficacy. These systems should be capable of accessing real-world medical knowledge and adopting evidence-based drug combinations. Third, while deep learning-based systems can generate treatment recommendations, their decision-making processes are often black-box, making it challenging for physicians and patients to understand the rationale behind suggested treatment plans. Furthermore, physicians should retain the ability to intervene and provide feedback to further optimize the AI system's decision-making.

To address these challenges, this paper aims to design and

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The source code are available at the following repository: <https://github.com/360ZMEM/PD-LLM>.

TABLE I
COMPARISON OF THIS WORK WITH EXISTING METHODS

Methods	Real-world data	Dosage data	Personalized treatment plans	Textual information	Knowledge base	Explainable treatment plans
Watts <i>et al.</i> [4]	✗	✗	✓	✗	✗	✗
Kim <i>et al.</i> [5]	✓	✗	✗	✗	✗	✓
Baucum <i>et al.</i> [6]	✓	✓	✓	✗	✗	✗
Riasi <i>et al.</i> [7]	✓	✓	✓	✗	✗	✗
Ours	✓	✓	✓	✓	✓	✓

validate a decision-making system for Parkinson’s Disease treatment based on large language models (LLMs). Specifically, we initially apply Monte Carlo Tree Search (MCTS) to identify optimal treatment strategies by simulating various treatment paths and their corresponding outcomes within a constrained search space. Following this, the LLM, acting as a “virtual physician,” utilizes both numerical and textual patient data, alongside the medication dataset generated by MCTS, to deliver personalized treatment recommendations. To further enhance the system’s capabilities, we integrate Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) reasoning techniques within the LLM system. This integration ensures that each treatment strategy is accompanied by detailed explanations and references to similar historical cases. Finally, we use the PPMI dataset, a real-world clinical database for PD, and incorporate medical guidelines to construct a local knowledge vector database that supports these processes. To assess the performance of this LLM-based approach, we conduct experiments comparing it against reinforcement learning (RL) methods and traditional physician regimens, evaluating its potential for improving decision-making in PD treatment. The main contributions of this work are as follows:

- **Development of a Novel Decision-Making System:** We propose the first decision-making system for PD treatment based on LLMs, which incorporate natural language inputs, including medical guidelines and patient records. This approach enhances the personalized treatment recommendations, improving the efficacy of PD management.
- **Integration of Advanced AI Techniques:** For **effectiveness**, we incorporate textual data and use numerical data in its raw form to maintain accuracy and integrity. We have developed a workflow that combines traditional data-driven methods with LLMs, using MCTS to identify optimal treatment plans and create a corresponding medication dataset. For **reliability**, we integrate RAG into the LLM framework, enabling the retrieval of relevant historical cases. To improve **interpretability**, the system applies CoT reasoning to provide treatment recommendations accompanied by detailed explanations, making decisions transparent. Additionally, this system can further engage in interactions with both physicians and patients, showcasing its potential for interactivity.
- **Exceptional Performance on Real-World Datasets:** Using the PPMI dataset, we conduct a comprehensive comparison of our LLM-based approach with traditional deep learning and physician methods. The results high-

light the potential of LLMs in advancing PD treatment strategies.

II. RELATED WORK

A. Data-Driven and Machine Learning-Based Parkinson’s Disease Treatment

Current AI-powered approaches to PD management primarily focus on diagnostic or detection tasks, with comparatively less emphasis on treatment. Treatment strategies, by contrast, often focus on developing personalized, fine-grained medication policies for individual PD patients. We have summarized recent methods for Parkinson’s treatment in Table I.

Watts *et al.* [4] proposed a Deep Reinforcement Learning (DRL)-based framework for optimizing medication dosage in Parkinson’s Disease. However, this approach relies on assumed patient data and has not been validated on real-world datasets. Kim *et al.* [5] classified patients into 28 discrete states using decision trees, subsequently applying a Markov Decision Process (MDP) to derive the optimal medication combination for each state, excluding medication dosages. While the simplification of states and policies contributes to some degree of interpretability, it fails to provide personalized medication strategies for individual patients. Baicum *et al.* [6] integrated DRL with wearable device datasets to propose a personalized, dynamically adjustable medication dosage adjustment method. Experimental results demonstrated that the RL-based approach outperformed physicians in treatment efficacy. Riasi *et al.* [7] employed a recurrent neural network (RNN) model to predict medication dosages, thus overcoming the limitations associated with the curse of dimensionality in RL action spaces. However, regression-based methods are confined to replicating physician strategies and lack the capability to explore or identify more optimal treatment solutions.

While these studies highlight the potential of AI in PD treatment, they also expose several critical limitations. First, the non-linear nature of neural networks often leads to a lack of interpretability, making it difficult to clearly explain the decision-making process. Furthermore, effective PD treatment necessitates the integration of patient history, medical guidelines, and physicians’ diagnostic opinions, all of which are textual data that play an integral role in clinical treatment. Neglecting these elements may result in suboptimal treatment outcomes. In conclusion, although efforts have been made to develop AI-assisted medication strategies for PD management, ensuring the effectiveness, reliability, and interpretability of these systems still remains a significant challenge.

B. Large Language Model Agent

Large language models (LLMs), trained autoregressively on vast amounts of natural language data [8], exhibit emergent phenomena that endow the models with general intelligence [9], enabling them to simulate human behavior. Concurrently, as technological advancements progress, LLMs have evolved beyond generating high-quality natural language text, now demonstrating expertise comparable to that of human specialists in fields such as medicine [10], [11], law [12], finance [13], marketing [14], and education [15], [16].

To further enhance the effectiveness of LLMs as human assistants, a range of techniques have been developed to improve their reasoning ability, reliability, and interpretability. RAG integrates LLMs with external knowledge bases [17], thereby improving the quality of responses in specialized domains without requiring additional training. It also allows for dynamic updates, contributing to improved reliability. The CoT [18] reasoning technique enhances model transparency by requiring the system to output a sequence of intermediate reasoning steps before reaching a final answer. This enables users to observe the model's cognitive process directly, rather than merely its final conclusion, thereby providing valuable insights for human-assisted decision-making [19]. These measures significantly enhance reliability and reduce the occurrence of hallucinations [20].

C. Applications of LLMs in Medicine and Parkinson's Disease

LLMs have exhibited significant potential in the medical field [21], [22]. Broadly, the applications of LLMs can be categorized into two primary categories: the first involves pre-training on extensive medical knowledge bases or performing In-Context Learning (ICL) within external medical knowledge environments. This approach enables the development of models capable of performing general tasks, such as medical record analysis, medical text summarization, and personalized healthcare assistance [23], [24]. The second category focuses on utilizing individual patients' numerical data, such as continuous monitoring data from wearable devices, diagnostic and prescription records from electronic health records, to conduct health status classification and disease prediction [11], [25]. These studies illustrate that by integrating the language comprehension capabilities of LLMs with expert knowledge from the medical domain, it is possible to enhance medical efficiency and diagnostic support to varying extents. In the domain of neurodegenerative diseases, particularly Parkinson's Disease (PD), researchers have begun investigating the potential of incorporating LLMs into disease management. For instance, Bouchouras *et al.* [26] developed a LLM-based model to assist with PD diagnosis and alert systems; Koga *et al.* [27] compared the diagnostic accuracy of ChatGPT and Google Bard in neurodegenerative pathology, though neither study addressed treatment regimens. Cardenas *et al.* [28] enhanced detection systems by integrating LLM chatbots, which provided interactive preventive and rehabilitation advice for patients, though they did not offer personalized medication dosage recommendations. Additional efforts have focused on automating the analysis of large-scale data, such as the work

by Stroganov *et al.* [29], who used LLMs to extract and standardize pathological reports from PD patients, significantly improving the efficiency of data analysis.

Overall, these efforts have laid a preliminary foundation for the application of LLMs in PD, but key aspects such as direct optimization decision of dosage and frequency of medications, personalized decision-making, model interpretability and reliability, and the iteration of medical knowledge remain insufficiently explored. Therefore, leveraging LLMs to provide interpretable, clinically relevant, and medication-specific decision-support systems in the context of complex diseases presents an urgent and important area for further research.

III. PRELIMINARY

A. PD Treatment

The current standard treatment for Parkinson's Disease is Levodopa (L-dopa), which facilitates dopamine replacement and is generally regarded as the most effective therapy for PD. While Levodopa effectively reduces bradykinesia (slowness of movement), it can exacerbate dyskinesia (involuntary movements). Therefore, effective Levodopa treatment requires a delicate balance of the contradictory effects of bradykinesia and dyskinesia.

Levodopa is available in several formulations: Immediate Release (IR), Controlled Release (CR), and Rytary (extended-release). IR is rapidly absorbed, providing quick symptom relief for acute exacerbations, but prolonged use may lead to dyskinesia. In contrast, CR is designed to prolong the release of the medication, resulting in a relatively slower onset of action, which may not be sufficient to promptly address acute flare-ups. Rytary combines both release mechanisms, providing fast relief initially, followed by sustained release to maintain the therapeutic effect. However, due to the complexity of Levodopa's release mechanism in Rytary, patients often experience fluctuations in drug efficacy, with periods of symptom relief followed by exacerbations.

As a result, optimizing Levodopa therapy for PD patients requires careful monitoring and drug adjustments. To address this challenge, there is a growing need for a data-driven framework to optimize medication management that can dynamically adjust treatment plans based on symptom tracking and individual patient profiles.

B. PPMI Dataset

The Parkinson's Progression Markers Initiative (PPMI) is a large-scale observational study on Parkinson's Disease, which has conducted long-term, periodic assessments of motor function and symptoms in over 700 patients. The follow-up intervals for these patients were every three or six months, during which they underwent motor assessments. All patients in the study received standard treatment with Levodopa. We selected 504 patients who exclusively used specific formulations of Levodopa (IR, CR, and Rytary). Table II presents the overview of the descriptive statistics for the PPMI dataset.

The PPMI dataset consists of three main components: Patient Covariate Data, Symptom Data (UPDRS scores), and Medication Data. The patient covariates include three key

TABLE II
SUMMARY OF PPMI PATIENT CHARACTERISTICS

	All Patients	L-Dopa Type		
		IR	CR	Rytary
N	504	482	152	64
Female (%)	37.9	37.9	41.4	39.1
Age	65±10	65	64	64
Years PD	1.4±1.9	1.4	1.4	2.0
Initial MDS-UPDRS-III	23.6±12.0	23.7	23.0	22.4
MDS-UPDRS-III Change	4.9±13.8	4.9	4.7	3.9
Daily LED (mg)	528±328	536	633	738

features: gender, age, and the number of years since the diagnosis of PD. The UPDRS scores, based on revised Unified Parkinson's Disease Rating Scale (MDS-UPDRS), are used to evaluate the severity of motor impairment in these patients. The medication dataset records each patient's PD medication regimen, including the type of medication (IR, CR, and Rytary), daily dosage, and the frequency of administration.

We cross-referenced the patients' covariates, symptoms, and medication records to generate time-series data corresponding to each patient's medication history.

C. Symptom Progression Prediction Model

Although Parkinson's disease treatment is a dynamic process without a definitive endpoint, we model it as a finite MDP with a discretized action space (limited medication regimens) and a finite number of medication cycles, enabling evaluation of treatment effects over a fixed time horizon. We model the patient's symptom progression as an MDP, defined as a tuple $\langle S, A, T \rangle$, where:

- S is the set of patient states, with $s_t \in S$ denoting the patient's symptom state at time t .
- A is the set of available treatment actions, with $a_t \in A$ denoting a certain medication combination taken at time t .
- T is a deterministic model that describes the patient's state transition. T maps the current state and action (s_t, a_t) to the next state s_{t+1} , with the patient's UPDRS score changing accordingly.

For the symptom progression prediction model, we adopt the MDS-UPDRS-III prediction model proposed by Baucum *et al.* [6]. Specifically, this model utilizes pharmacokinetic modeling to convert daily medication dosage and frequency into a 35-dimensional vector. This vector quantifies the blood drug concentrations measured at 30-minute intervals throughout 17 waking hours. Combined with regularized patient covariates and past UPDRS scores, the model predicts the patient's UPDRS score for the next visit. Since both the input and output in the MDS-UPDRS-III model require normalization, we scale the patients' UPDRS scores to fit within a normalized range of 0 to 53, reflecting that 53 corresponds to the 99th percentile of the actual UPDRS score distribution.

This progression prediction model can be mathematically expressed as follows:

$$x_t = x_{t-1} + \hat{\Delta}x_t + \epsilon$$

$$\hat{\Delta}x_t = f([\Delta x_{(t-t_{\text{past}}):(t-1)}, \ell_{t-1}, \mathbf{c}]^T)$$

where:

- x_t is the (arctanh-transformed) UPDRS score at time t .
- $\Delta x_t = x_t - x_{t-1}$ denotes the change in the score, i.e., the difference between the scores at time t and $t-1$.
- $\hat{\Delta}x_t$ represents the predicted change in the score.
- ϵ is a Gaussian disturbance term that follows a normal distribution with a mean of 0 and standard deviation σ .
- ℓ_{t-1} is the L-dopa profile of the patient at time $t-1$, which quantifies the concentration levels of Levodopa in the blood.
- $\mathbf{c}$ is a vector of covariates, including gender, age, years since PD diagnosis.

IV. METHODOLOGY

In this section, we propose a framework named **PD-LLM**, designed to use LLMs to provide personalized medication recommendations for PD patients. As illustrated in Figure 1, the PD-LLM framework consists of three key components:

- **Finite-Horizon MCTS**: Generation of the probabilistically optimal medication recommendation dataset (Subsection A).
- **Similar Case Retrieval**: Knowledge retrieval based on vector similarity for retrieving clinically relevant cases (Subsection B).
- **CoT Reasoning**: Step-by-step reasoning to improve the interpretability of medication strategies (Subsection C).

The workflow begins with textually processing original patient cases, converting clinical data and medication records into formats suitable for LLM processing. Using the Finite-Horizon MCTS method, we leverage the symptom prediction model to simulate disease progression and identify optimal medication strategies. These strategies are compiled into a medication recommendation dataset and incorporated into the vector knowledge base of the RAG system. When a patient seeks treatment, the system calculates vector similarity to retrieve relevant cases from the knowledge base, which include recommended treatment plans. The input for the LLM consists of three key components: textually processed patient records, similar cases, and authoritative Parkinson's disease guidelines. This retrieved information is integrated into the prompt alongside CoT reasoning, enabling the LLM to perform step-by-step analysis and generate comprehensive medication recommendations accompanied by detailed explanations. The PD-LLM framework is illustrated in Figure 1.

A. Finite-Horizon MCTS

Before integrating LLMs into medication decision-making, we employ machine learning-based prediction methods to support and refine the process. Although LLMs utilize extensive prior knowledge to provide empirically accurate recommendations, relying solely on experience-based suggestions may lead to errors. To address this, we incorporate MCTS [30], [31], a probabilistic method for optimizing solutions in finite decision spaces with predefined endpoints through limited simulations.

Specifically, we construct a tree structure in which the root node represents the patient's current state. During each

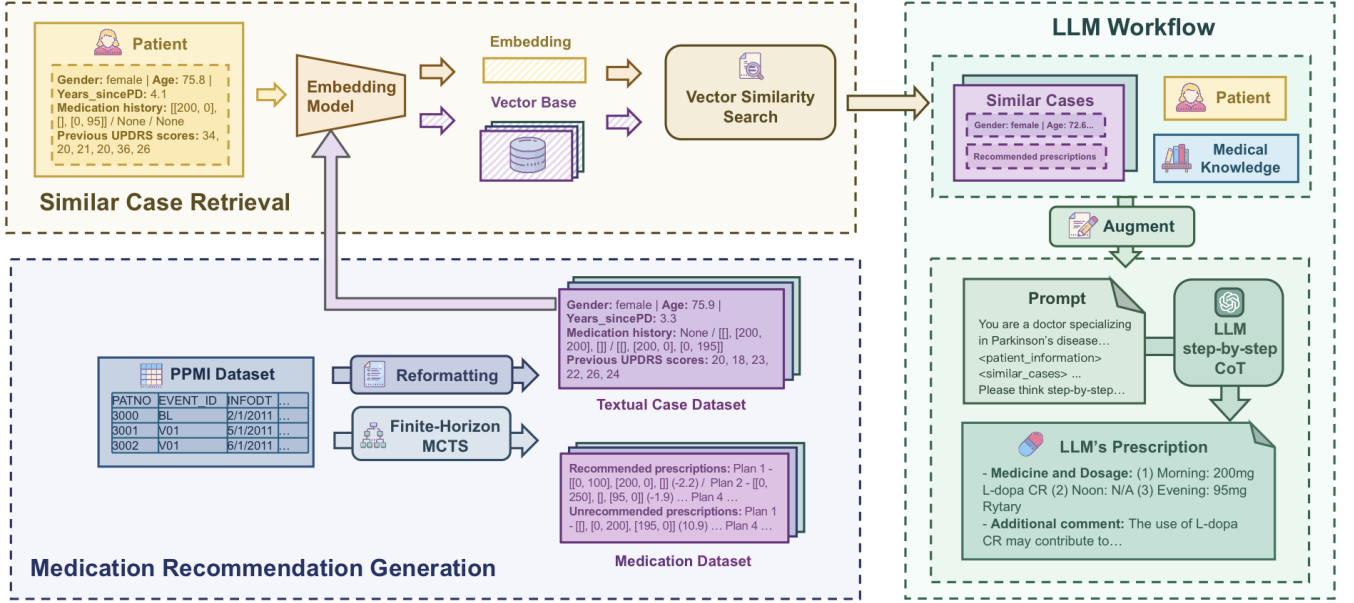


Fig. 1. The overall architecture of PD-LLM framework.

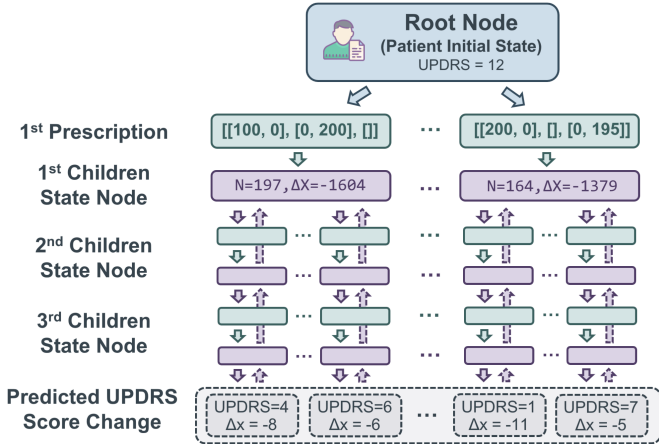


Fig. 2. A simple illustration of Finite-Horizon MCTS simulation and back propagation process.

simulation, a medication combination is selected from the action space (details of the action space implementation are provided in Appendix B). The MDS-UPDRS-III model is then used to predict the patient's UPDRS score at the next follow-up visit under the selected plan. The simulation transitions to a new state, represented as children node of the current root, and the process continues recursively to simulate three consecutive medication administrations.

Based on the patient's current state information, we conduct randomized simulations of medication strategies. To evaluate both the short-term and long-term effects of each medication plan, we use the expected UPDRS score change after three follow-up visits as the evaluation criterion. After completing these iterations, the patient's UPDRS score change, relative to their initial state, is calculated and back-propagated through the tree structure. Figure 2 illustrates the simulation process,

and Algorithm 1 presents the corresponding pseudocode. Notably, since each medication recommendation leads to a unique patient state, we associate each action with its resulting state in Algorithm 1. This approach extends the traditional Upper Confidence Threshold (UCT)-based MCTS into a finite-horizon version, tailored specifically for Parkinson's disease treatment.

The priority score of each action during single-step plan selection is determined by the following:

$$UCT|_{s' \rightarrow T(\mathbf{LSTM}, s, a)} = -\frac{\Delta X(s')}{N(s')} + C_p \sqrt{\frac{2 \ln N(s)}{N(s')}}$$

Where:

- $N(s)$ and $N(s')$ are counters that record the number of times the current state node and its children state node have been reached in previous simulations, respectively.
- $\frac{\Delta X(s')}{N(s')}$ is the expected UPDRS score change at the children state node.
- $\mathbf{LSTM}$ represents the MDS-UPDRS-III predict model.
- C_p is a hyperparameter balancing exploration and exploitation.

To ensure sufficient exploration of all nodes, the UCT value is set to positive infinity for actions leading to child nodes with $N(s') < 10$. This mechanism prioritizes under-visited child nodes for random selection and further exploration, ensuring sufficient exploration across the action space.

However, the MCTS approach has several limitations. Specifically, MCTS requires extensive simulations for each single-step decision, repeatedly invoking the prediction model to estimate the expected UPDRS scores. This not only incurs significant computational costs but also makes the algorithm highly dependent on the accuracy of the prediction model, and may introduce cumulative errors. Moreover, MCTS lacks

Algorithm 1 Finite-horizon MCTS for PD prescription searching

Require: Root node s_0 , action space A , MDS-UPDRS-III Predict Model **LSTM**

```

1:
2: function BACKPROPAGATE(Node  $s$ , score change  $\Delta x$ )
3:   while  $s$  is not None do
4:      $N(s) \leftarrow N(s) + 1$ 
5:      $\Delta X(s) \leftarrow \Delta X(s) + \Delta x$ 
6:      $s \leftarrow \text{parent of } s$ 
7:   end while
8: end function
9:
10: function GETBESTCHILD(Node  $s$ )
11:   return  $\text{argmax}_{s' \leftarrow T(\text{LSTM}, s, a), a \in A} -\Delta X(s')/N(s') +$ 
         $C_p * \sqrt{2 * \ln N(s)/(N(s') * \text{bool}(N(s') > 10))}$ 
12: end function
13:
14: function EXPANDNODE
15:   for  $t = 0 \rightarrow t = 2$  do
16:      $s_{t+1} \leftarrow \text{GetBestChild}(s_t)$ 
17:   end for
18: end function
19:
20: for Number of simulations do ▷ Main Process
21:    $s_3 \leftarrow \text{ExpandNode}(s_0)$ 
22:   Obtain the UPDRS score difference  $\Delta x$  between  $s_3$ 
   and  $s_0$ 
23:   BackPropagate( $s_3$ )
24: end for
25: Evaluate an action based on the value of  $\Delta X/N(s_1)$  of
   its corresponding transition state, for any  $s_1 \in \text{children of}$ 
    $s_0$ 

```

generalizability, as it cannot share treatment experiences across different patients. Its design is primarily suited to discrete action spaces, making it challenging to handle continuous action spaces, which limits its applicability in designing flexible medication regimens.

To address these challenges, we propose a workflow that combines MCTS with RAG. The medication recommendation dataset generated by MCTS is incorporated into a vector knowledge base and serves as part of the input to the LLM. This integration enables the LLM to generate personalized and optimized medication recommendations.

B. Vector Knowledge Base and Similarity Search

1) Structuring Patient Information and the Medication Dataset: We use concise and highly readable natural language to describe patient information, including key elements such as age, gender, clinical history, and medication history. Each case is simulated using Finite-Horizon MCTS. Based on the expected UPDRS score change, the search space is filtered to extract the four most recommended and four least recommended medication regimens, which form the medication recommendation dataset. Each regimen is represented in a structured

Python list format, accompanied by its corresponding expected UPDRS score. This structured format enables LLMs to analyze medication regimens, accounting for factors such as dosage and frequency. By offering multiple alternative regimens, this approach enhances the model's ability to evaluate options and generates more precise, personalized, and clinically informed treatment recommendations.

2) Similar Case Retrieval Based on Text Embedding Similarity: To complete the RAG workflow, we identify the most relevant similar cases and their corresponding treatment recommendations by evaluating the similarity between a given patient and other cases in the dataset. The detailed steps are as follows:

Text Embedding Generation. A pre-trained text embedding model is used to encode the target patient's information into an embedding vector, denoted as emb_p . Similarly, the information of all cases in the dataset is encoded to construct a vector knowledge base, represented as emb_b .

Similarity Calculation. Cosine similarity is employed to measure the similarity between the target patient's embedding vector emb_p and the embedding vectors of all cases emb_b . The similarity is calculated using the following formula:

$$\text{similarity} = \frac{\text{emb}_p \cdot \text{emb}_b}{\|\text{emb}_p\| \|\text{emb}_b\|}$$

The resulting similarity score ranges between 0 and 1, where higher scores indicate greater similarity.

Selection of Similar Cases. To ensure accurate matching, we utilize only the structured patient information (e.g., demographic details) during the encoding process. This ensures the encoded similar cases align with the prescribed patient's structured information, excluding unstructured data like medication regimens. From the dataset, the two cases with the highest similarity scores are identified and their corresponding treatment information is retrieved. These selected cases are then incorporated into the input prompt for the LLM, serving as references for making medication decisions.

By adopting this approach, we integrate the target patient's individual characteristics with similar cases. This method leverages vector similarity retrieval to provide high-quality reference information for the LLM, thereby significantly improving the precision and personalization of its medication recommendations.

C. Prompt Design

The input prompt incorporates key information such as patient details, data from similar cases, recommended medications, and relevant medical knowledge. To address task-specific requirements, we developed a step-by-step, domain-knowledge-driven prompt. This prompt leverages the LLM's semantic understanding to synthesize the patient's condition with similar case data, and follow a clear reasoning process to generate outputs [32]. Figure 3 illustrates a simplified version of the prompt, with the full version provided in the Appendix D. The prompt design includes the following components:

Structured Descriptions for Patient and Similar Case Information. The prompt includes concise and structured

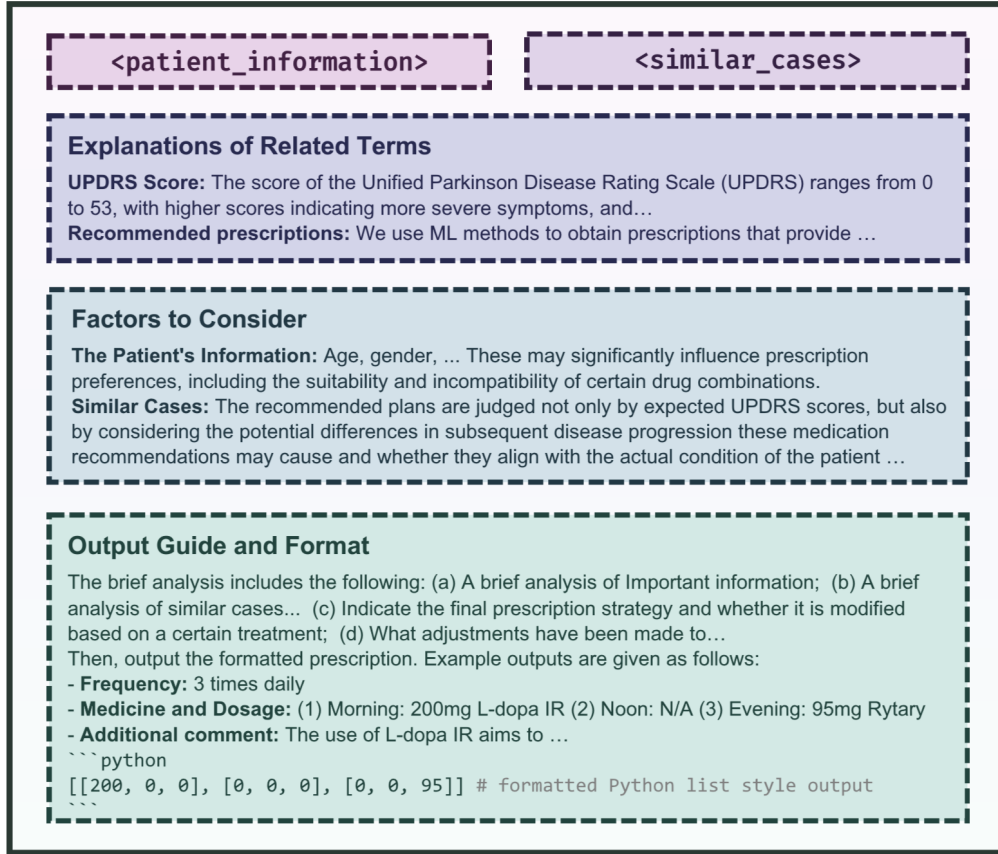


Fig. 3. Simplified prompt example used in the PD-LLM framework.

natural language descriptions of the target patient and similar cases, along with explanations of their potential influence on treatment. These prompts help the LLM understand the contextual background and apply relevant medical knowledge effectively.

Step-by-Step Chain-of-Thought Reasoning. The LLM analyzes patient information, examines similar cases and their suggested regimens, and verifies their relevance. Finally, it balances multiple factors to provide a comprehensive explanation and generate tailored medication recommendations.

Structured Output for Medication Plans. The medication plans generated by the LLM are presented in a textual format, which includes frequency, dosage, as well as a clear explanation of the rationale behind the plan's design. The output is then converted into a structured Python list format to facilitate automated processing and ensure clarity.

Overall, this framework integrates patient-specific and similar-case data with explicit reasoning and professional expertise. By guiding the LLM through a systematic reasoning process and stepwise outputs, it ensures a complete reasoning chain and provides personalized, reliable, and interpretable medication recommendations.

V. EXPERIMENT

A. Data Processing

The original PPMI dataset contains 504 patients who used specific formulations of Levodopa, with each patient under-

going k follow-up visits. Due to variations in the number of follow-ups across patients, we retained only those patients with $k \geq 7$. This filtering resulted in a final cohort of 399 patients, representing approximately 79% of the original dataset. The average interval between treatments for these patients ranged from 4 to 12 months. Additional statistical information is provided in Appendix A.

To make the most of limited data while preventing data leakage, we employed Leave-One-Out Cross-Validation (LOOCV), a specific form of k -fold cross-validation where k equals the total number of patients in the dataset. During the testing phase, the PD-LLM framework generates medication strategies for each individual patient, utilizing a similar-case database that exclusively contains historical data from the remaining 398 patients.

B. Implementation Details

In implementing the PD-LLM workflow, we focused on the following key aspects:

LLM Selection and Parameter Configuration: We selected the DeepSeek-v2.5 model [33] as the base LLM, with parameters set to `temperature = 0.2` and `Top P = 1` to ensure deterministic and consistent text generation. This selection was based on DeepSeek-v2.5's exceptional performance on widely recognized benchmarks for large language models and its cost-efficiency with low cost per million tokens.

MCTS Configuration: The MCTS action space comprises 73 medication regimens, each defined by a specific combination of drug types, dosages, and dosing frequencies (provided in Appendix B). The dataset included 399 patients, contributing a total of 2,000 unique states, as patient states varied across follow-up visits. For each patient state, MCTS simulations were employed to predict the expected change in UPDRS scores after 3 follow-up visits, optimizing treatment strategies to construct the medication recommendations dataset. Each state underwent 2,500 simulations, with the number of simulations determined through a convergence analysis of the Monte Carlo method to ensure stability and reliability. The robustness analysis of MCTS simulation can be found in Appendix C.

Similar Case Matching: We employed the MedEmbed-large embedding model [34] for similar case matching to ensure relevance and accuracy in the retrieved results. MedEmbed-large has demonstrated superior performance in medical text embedding tasks, effectively capturing the semantic similarity between cases. During the matching process, records from the same patient (identified by the same PATNO) were excluded to prevent data leakage and overfitting.

Prediction and Evaluation: We used the MDS-UPDRS-III model to predict the expected change in UPDRS scores and evaluate the long-term therapeutic outcomes of different medication regimens. The MDS-UPDRS-III model has been rigorously validated and is recognized for its high predictive accuracy.

C. Evaluation Metrics

This study employs the following metrics to evaluate treatment effectiveness. The first metric is **UPDRS Change**. The UPDRS Change is defined as the average change in UPDRS scores across all patients and is mathematically represented as:

$$\Delta U = \frac{1}{N} \sum_{i=1}^N \left(\text{UPDRS}_{k_i}^{(i)} - \text{UPDRS}_0^{(i)} \right)$$

where:

- N is the total number of patients included in the study.
- $\text{UPDRS}_0^{(i)}$ is the baseline UPDRS score at time point $t = 0$ of patient i .
- $\text{UPDRS}_{k_i}^{(i)}$ is the UPDRS score of patient i at their last follow-up visit k_i .

For each patient, a new medication strategy is implemented after each follow-up visit, and the MDS-UPDRS-III model predicts the UPDRS score at the next visit. This process continues until k follow-up visits to calculate the UPDRS score difference through the whole process. This allows for a direct comparison of UPDRS Change (ΔU) across the proposed medication strategies. Statistical metrics such as mean, median, 25th percentile, and 75th percentile of the UPDRS Change are used for evaluation. ΔU serves as a “lower is better” metric because the UPDRS is designed to assess the severity of symptoms of a patient, with higher scores indicating worse motor impairments or more severe symptoms.

In addition to the UPDRS score, we also employed two metrics: **the proportion of improvement** and **significant im-**

TABLE III
KEY METRICS FOR UPDRS CHANGE (ΔU) UNDER DIFFERENT EXPERIMENTAL SETTINGS.

Settings	Average↓	Median↓	25th percentile↓	75th percentile↓
Real	5.17	4.00	-4.00	14.00
Physician	3.26	3.46	-2.98	10.06
RL	2.84	3.15	-2.05	8.72
PD-LLM	1.83	1.86	-3.83	8.14

provement. Specifically, for each individual patient, we evaluated the difference in ΔU between the physician-administered treatment and the method under evaluation. If the ΔU of the evaluated method was smaller, it was considered to provide a therapeutic improvement. Furthermore, if the ΔU of the evaluated method decreased by more than 2 points compared to the physician-administered treatment, it was deemed a significant improvement. Finally, we calculated the percentage of patients meeting either criterion among the total number of patients evaluated [6].

D. Baselines

To evaluate the effectiveness of our proposed framework, we designed the following three baseline approaches for comparison.

Real: The actual UPDRS score change were recorded in the PPMI dataset following physician-administered treatment.

Physician-method: This method uses the prescriptions provided by physicians in the PPMI dataset, with change in UPDRS scores predicted using the MDS-UPDRS-III model. It represents current clinical practices and serves as a highly relevant benchmark for real-world reference. While discrepancies between the Physician-Method and real outcomes are inevitable due to inherent prediction errors in the MDS-UPDRS-III model, the UPDRS change predicted by the Physician-method remain comparable to those of the RL and PD-LLM methods.

RL-method: This baseline employs the Reinforcement Learning (RL) model proposed by Baucum et al. [6], which generates medication recommendations by leveraging an action space of potential treatment combinations. A ten-fold cross-validation process is applied to the dataset of 399 patients to prevent data leakage. The model is rigorously trained and validated to explore long-term treatment strategies. Details of the RL model’s configuration is provided in Appendix B.

These baseline methods represent both current clinical practices and advanced data-driven AI approaches, enabling an effective comparison to demonstrate the superiority of our proposed PD-LLM framework in optimizing personalized long-term treatment outcomes for PD patients.

E. Main Results

Table III presents four metrics of UPDRS Change (ΔU) under different experimental settings. The results indicate that the PD-LLM approach achieves an average improvement of 1.43 points in ΔU compared to the Physician group, whereas the RL method shows a smaller improvement of 0.42

TABLE IV
ABLATION STUDY: KEY METRICS OF UPDRS CHANGE (ΔU) FOR
PD-LLM WORKFLOW COMPONENTS.

Metric	Average↓	Median ↓	25th percentile ↓	75th percentile ↓
Physician	3.26	3.46	-2.98	10.06
w/o RAG	2.68	3.09	-3.52	9.04
w/o MCTS	2.71	2.75	-2.99	8.50
w/o VSS	2.40	3.00	-3.50	8.71
w/o CoT	2.26	2.80	-3.42	9.03
PD-LLM	1.83	1.86	-3.83	8.14

points relative to the Physician group. Furthermore, PD-LLM outperforms across all statistical metrics, indicating significant overall improvements. A t-test analysis confirms that the improvement by the RL method over the Physician group is statistically significant ($p = 0.029$), while the improvement achieved by the PD-LLM approach is highly significant ($p = 3.2 \times 10^{-9}$).

F. Ablation Study

To validate the effectiveness of the PD-LLM workflow and assess the contribution of each component, we designed four ablation experiments. The primary components of PD-LLM include RAG and CoT. Within RAG, two critical subcomponents are MCTS and Vector Similarity Search (VSS).

The four ablation experiments are designed as follows:

- Without RAG (**w/o RAG**): This configuration removes all RAG components, requiring the LLM to directly analyze patient information using CoT reasoning to provide medication strategies and explanations. This experiment evaluates the overall impact of RAG on performance.
- Without MCTS (**w/o MCTS**): This configuration retains VSS but replaces the MCTS medication plans with recommendations generated by the RL policy network, aiming to assess the contribution of MCTS.
- Without VSS (**w/o VSS**): This configuration retains MCTS but replaces VSS with random sampling of reference cases from the medication recommendation dataset. This evaluates the contribution of VSS.
- Without CoT (**w/o CoT**): This configuration retains all RAG components but removes the CoT prompts. The LLM provides only medication dosages without reasoning or explanations. This experiment assesses the impact of CoT on the overall workflow.

To ensure the robustness of the ablation study, we performed 100 repetitions of the k follow-up visits for each patient. This thorough evaluation confirms the importance of each component in PD-LLM and establishes a clear hierarchy of their contributions. These results provide valuable insights for future optimization and extension of PD-LLM, ensuring both the reliability and consistency of its performance.

Table IV presents the variations in UPDRS Change across different ablation groups, emphasizing the importance of each component in the PD-LLM framework. RAG emerges as the most critical component of PD-LLM, playing a pivotal role in integrating external knowledge and optimizing medication

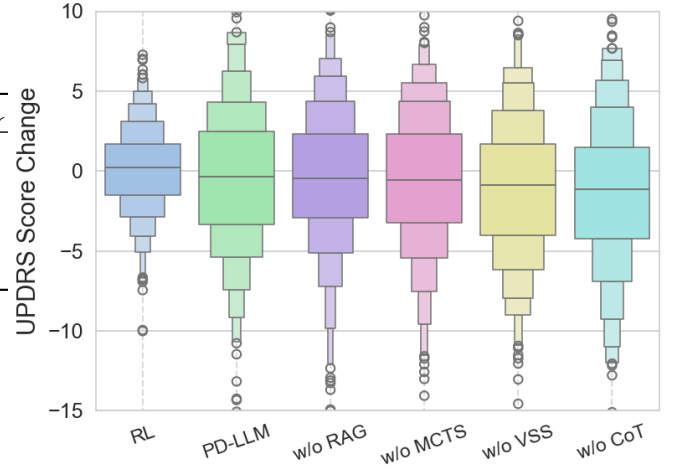


Fig. 4. Enhanced box plot of UPDRS score change for RL and various LLM workflows vs. Physician setting.

TABLE V
METRICS OF TREATMENT EFFICACY IMPROVEMENTS FOR RL AND
VARIOUS LLM WORKFLOWS VS. PHYSICIAN SETTING.

Metrics	Improvement (%)↑	Significant improvement (%) ↑
RL	52.6	24.3
w/o RAG	53.4	34.8
w/o MCTS	54.6	34.3
w/o VSS	54.4	35.3
w/o CoT	60.1	39.1
PD-LLM	61.7	43.4

strategies. Its removal results in the most significant performance decline. Among the RAG subcomponents, MCTS and VSS contribute significantly to improving treatment outcomes by optimizing medication plans and ensuring the relevance of reference cases, respectively. Although CoT has a relatively smaller impact on UPDRS Change, its ability to explain medication recommendations enhances user understanding and trust in the AI system, thereby indirectly contributing to better treatment results.

Additionally, Table V shows the metrics of treatment efficacy improvements when compared to the Physician setting, and Figure 4 shows the corresponding enhanced box plot of UPDRS score change. It can be seen from Table V that our PD-LLM achieves a 61.7% improvement and a 43.4% significant improvement, which remarkably outperforms the RL setting mentioned above.

VI. DISCUSSION

A. Evaluation and Comparison of Treatment Regimens

To evaluate medication preference differences among the PD-LLM, Physician, and RL methods, we conducted a comparative analysis focusing on the proportions and daily dosages of medications in treatment regimens, as presented in Table VI. And Figure 5 illustrates the dosage distribution for each method.

1) *Medication Proportions and Dosage*: In terms of medication proportions, the Physician group predominantly relied

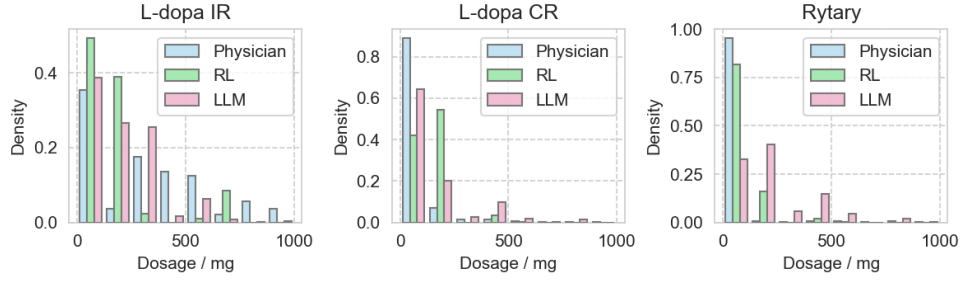


Fig. 5. Histogram distribution of Levodopa dosages under different methods.

TABLE VI
SUMMARY OF MEDICATION USAGE UNDER DIFFERENT EXPERIMENTAL SETTINGS.

	Physician	RL	LLM
Taking IR (%)	66.7	78.8	72.9
Taking CR (%)	14.7	72.5	51.1
Taking Rytary (%)	4.7	31.2	86.3
IR daily dose (mg)	556	224	244
CR daily dose (mg)	216	189	239
Rytary daily dose (mg)	948	167	246
Doses per day	3.29	3.03	3.05

on L-dopa IR (66.7%), with much lower usage of L-dopa CR (14.7%) and minimal use of Rytary (4.7%). Furthermore, in approximately 20% of follow-up records, physicians opted not to prescribe any medication, reflecting a more conservative treatment approach. And they typically favored single-drug regimens. In contrast, both the RL and PD-LLM methods demonstrated greater ability to implement combination therapies that leverage multiple medications to optimize treatment outcomes. Notably, the PD-LLM group increased the usage of Rytary (86.3%), far surpassing the Physician group (4.7%) and the RL group (31.2%).

Regarding medication dosages, the Physician group administered the highest average daily dosages across patients. In contrast, due to its limited discrete action space, the RL group's dosage distributions appeared as isolated spikes on the histogram, a consequence of the few action options available. It primarily used a combination of L-dopa IR and CR, occasionally supplemented with high-dose IR or low-dose Rytary. By comparison, the PD-LLM offers greater flexibility in dosage adjustments, dynamically tailoring medication proportions to optimize outcomes based on patient needs, resulting in a more continuous histogram distribution. It allows for more precise dose adjustments, enhancing the personalization and accuracy of treatment strategies.

2) *Discussion and Insights*: The PD-LLM framework notably increased the use of Rytary, whose extended-release formulation provides a smoother pharmacokinetic profile. This enhancement minimizes fluctuations in drug efficacy, ensuring more stable therapeutic effects. However, its use requires meticulous dose adjustments and careful monitoring. It is worth noting that Rytary was only approved by the FDA in 2015, which may explain its limited adoption in physician-prescribed regimens.

In addition, PD-LLM demonstrated remarkable flexibility in personalized combination therapies by using L-dopa IR or CR during the daytime and supplementing with Rytary at night. This tailored approach not only optimizes the overall therapeutic efficacy but also improves patients' quality of life, as further detailed in the case study in Section V.

Overall, PD-LLM exhibited distinct advantages in facilitating combination therapies, enabling flexible dosage adjustments, and designing personalized treatment regimens, thereby offering promising advancements for Parkinson's Disease management.

B. Case Study

To provide a more detailed and intuitive understanding of the medication strategies devised by PD-LLM and its differences compared to the Physician and RL groups, we conducted a case study on two representative patients from the PPMI dataset, identified as 3050 (PATNO 3050) and 3057 (PATNO 3057). Figure 6 outlines the key demographic and daily prescriptions for these two patients. The daily medication recommendations provided by RL and LLM contain the medication type, dose, and schedule at different disease progression phases.

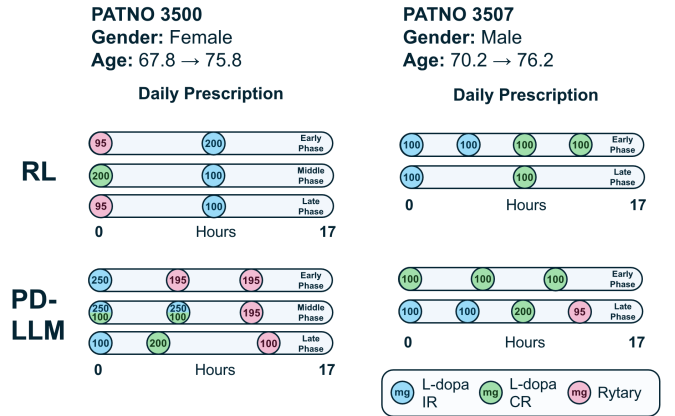


Fig. 6. Basic information and daily prescriptions given by RL and PD-LLM for two representative patients, including medication type, dosage, and schedule over a 17-hour day (from 5:00 am to 10:00 pm).

Figure 7 illustrates the predicted trends in UPDRS scores under different methods. As shown, Patient 3050 demonstrated stable disease control over time, with PD-LLM achieving

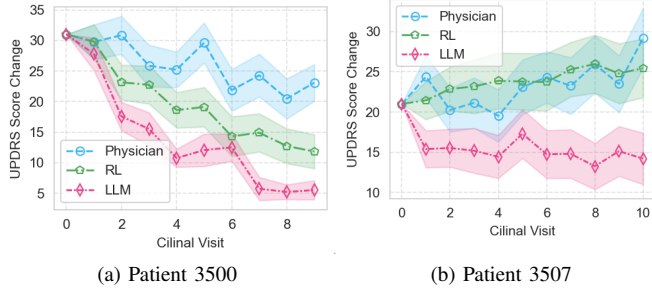


Fig. 7. Trends of UPDRS scores over time for two representative patients under different experimental settings.

better therapeutic outcomes compared to other methods. Conversely, Patient 3057, who exhibited a worsening disease trajectory, showed improvement with PD-LLM’s more effective symptom progression mitigation.

Figure 8 shows the corresponding daily L-dopa profiles for both patients, charting the blood concentrations of L-dopa throughout the day (from 5:00 am to 10:00 pm). It details the profiles for patient 3500 at the early (2nd clinical visit), middle (6th clinical visit), and late phase (9th clinical visit), as well as for patient 3507 at the early (1st clinical visit) and late phase (6th clinical visit). Additionally, we provide the complete LLM responses in Appendix E for further analysis of similar cases and the reasoning behind LLM’s outputs.

It can be observed that for patient 3500, during the early and middle stages, due to the patient’s poor condition, LLM increased L-dopa blood concentrations by administering higher doses of L-dopa IR and CR in the morning. In the late stage, as the patient’s condition improved, the dosages were appropriately reduced. Throughout the treatment process, LLM consistently provided Rytary in the afternoon to maintain stable blood drug levels and ensure continuous therapeutic effects. For patient 3507, whose initial UPDRS score was low, LLM administered continuous low doses of L-dopa. In the late stage, with the onset of fluctuations and a tendency for deterioration, LLM increased the dosage of L-dopa CR and supplemented it with Rytary to stabilize the blood drug concentration.

These case studies highlight the adaptability and efficacy of PD-LLM in managing varying disease stages and patient-specific needs, further underscoring its potential to improve personalized Parkinson’s Disease management.

C. Significance in practical applications

The proposed framework demonstrates significant practical value in clinical settings, particularly through its focus on interpretability and similarity analysis. First, the incorporation of Chain-of-Thought (CoT) reasoning addresses critical challenges related to clinical trust and communication. Medical decisions directly impact patient safety, CoT-generated explanations enable physicians to quickly evaluate the validity of the model’s decisions, avoiding blind reliance on “black-box” outputs, and facilitating effective communication with patients regarding treatment plans. Additionally, CoT reasoning reveals potential flaws in the model’s decision-making

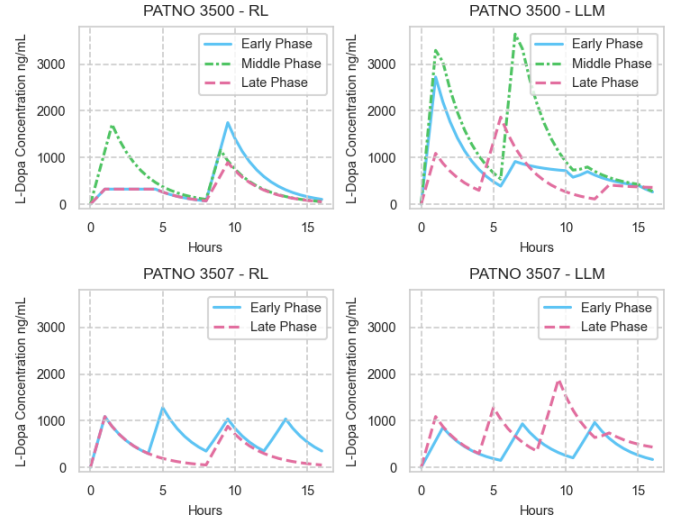


Fig. 8. Daily L-dopa profile at different clinic visit for two representative patients under different experimental settings.

process, allowing researchers to trace and rectify errors. This step-by-step reasoning not only enhances interpretability but has also demonstrated performance improvements in models such as GPT-o1 and Deepseek-R1. Furthermore, in high-risk domains like healthcare, interpretability ensures compliance with ethical and legal standards by making the decision-making process transparent, thereby mitigating risks of bias, discrimination, or legal violations.

Second, similarity analysis adds value by improving the quality of treatment strategies and optimizing system efficiency. By retrieving cases similar to the current patient’s profile from a large case database, the model leverages evidence-based treatment plans that have already proven effective. This reduces the risk of AI hallucinations and increases the relevance and effectiveness of the proposed strategies, leading to improved patient outcomes and higher-quality recommendations. Moreover, similarity analysis minimizes training costs by relying on a continually expanding and updated clinical case database. Instead of requiring frequent retraining or adjustments to the model’s core architecture, the system integrates the latest clinical data and research findings by simply updating the database. This dynamic integration ensures that the model remains current with evolving medical knowledge while maintaining operational efficiency.

Together, CoT reasoning and similarity analysis not only enhance the system’s performance and adaptability but also underscore its potential to transform personalized medical care in real-world applications.

VII. CONCLUSION

This study pioneers the integration of LLMs into personalized medication strategies for Parkinson’s disease, demonstrating significantly greater effectiveness, reliability, and interpretability compared to previous methods based on traditional machine learning and deep learning. Experiments using real-world datasets indicate that our decision system enables significant therapeutic improvements in 43.4% of patients

compared to physicians, and RL-based strategies achieves only 24.3% similar outcomes. These results strongly support the integration of LLM-driven decision-making into clinical treatments for Parkinson's disease. Despite these encouraging outcomes, further validation is necessary in larger clinical settings to address challenges arising from heterogeneous patient conditions.

Given the prolonged and complex nature of Parkinson's disease, future work could explore multi-agent collaborative strategies. This approach would allow multiple LLM agents to assume the roles of various multidisciplinary specialists, such as neurologists, speech therapists, rehabilitation experts, and dietitians. Together, they would collaboratively determine and dynamically adjust treatment plans, aiming to provide patients with more comprehensive, effective, and intelligent disease management services. Moreover, this multi-agent framework promises to enhance the outcomes and scalability of personalized medicine and open new avenues for research and practice in the comprehensive management of neurodegenerative diseases.

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